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DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

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DATE OF SUBMISSION:19/04/24

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Java EXP # 12

* AIM: To implement exceptions in Java (read input data Data Input Stream / Buffer Reader classes).

* Description:-

1) Difference between error and exception.

Error	Exception
1) Its an action that is inaccurate or Incorrect.	1) It is used to handle errors that can be recovered from within the program.
2) Cannot be handled or caught by the program.	2) Can be caught and handled in the program to maintain flow or recover the situation.
3) Syntax Errors, Runtime errors, logical errors.	3) Checked and unchecked exception.
4) May cause the program to terminate.	4) Disrupts the flow but allow for reliset or handling within the program.

2) Syntax of creating objects using Data Inputs Stream/Buffered Reader classes and their methods.

→ Data Input Stream

Syntax:-

```
Data Input Stream object-name = new Data Input Stream;
    object-name.read Byte();
    object-name.read Datatype();
```

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Buffer Reader:-

Syntax:-

```
InputStreamReader obj-name = new InputStream Reader();
BufferReader obj-name-2 = new BufferReader(obj-name);
obj-name 2.read datatype();
```

3) Types of exception :-

→ There are 2 types of exception:-
① Built-in Exception, ② User-Defined exception.

1) Built-in Exception:-
These exceptions are able to define the error situation so that we can understand the reason of getting this error. It has 2 types of exceptions i.e. checked and unchecked exception.

⇒ The checked exception are with compile time exception because these exception are checked at compile time by the compiler.

⇒ The unchecked exception are just opposite to checked exception. The compiler will not check these exceptions at compile time. In simple words, if a program throws an unchecked exception and even if we didn't handle or declare it, the program would not give a compilation error.

2) User-Defined exception
We already have some built-in exception. These exceptions are restricted to trigger on some predefined condition.

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~~We already have some built-in exception.~~
We can write our own exception class by extend the exception class.

4) Throwable class (Exception hierarchy)
→ The top of the exception class hierarchy is the Throwable class, which is a direct subclass of the object class. Throwable has 2 direct subclasses - Exception and error. The exception class is used for exception conditions that the application may need to handle. The error class is used to indicates a more serious problem in the architecture and should not be handled in the application code

5) Try, catch, Throw, Throws, and finally blocks.
① Try
It contains a set of statements where an exception can occur.
try {
// statement that might caught exception
}

② Catch:- Catch block is used to handle the uncertain condition of a try block.
Catch (Except^n name obj-name) {
// statements
}

3) Throws:- It is used to transfer from try block to the catch block.

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4) Throws :- It is used for exception handling without try and catch block.

5) Finally :-
It is executed after the catch block. We use it to put some common code. When the are multiple catch blocks.

12) b) Write a Java Program to create a user defined Exception class MarksOutofBoundsException. If entered marks of any subject is greater than 100 or less than 0, and then program should create a user defined Exception of type MarksOutofBoundsException and must have a provision to handle it.

Program :-

```
import java.util.*;
class MarksOutsOfBoundsException extend Exception {
    public MarksOutofBoundsException (int marks) {
        if (marks < 0) {
            System.out.println (" The subject marks
                                should be greater than equal to 0");

        if (marks >100) {
        System.out.println (" The subject marks should be less than
                            100"); }
    }
    String Point () {
        return "Marks Out of Bounds Exception";
```

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```
}
}

class StudentDetails {
    string name;
    int sub1, sub2, sub3 roll no;
    Scanner sc=new Scanner (System.in);
    public void getdetails () Throws MarksOutofBoundsException {
        System.out.println ("Enter Student name");
        name=sc.next();
        System.out.println ("Enter Sub1,sub2,sub3,marks");
        sub1 =sc.nextInt()
        sub2 =sc.nextInt();
        sub3 =sc.nextInt();

    if (sub1<0 || sub1>100) {
        throw new MarksOutofBoundsException (sub1); }
    else if (sub2<0 || sub2>100) {
        throw new MarksOutofBoundsException (sub2); }
    else if (sub3<0 || sub3>100) {
        throw new MarksOutofBoundsException (sub3); }
    else {
        display(); }
    }
    public void display () {
    System.out.println ("Student name : " + name);
    System.out.println ("Marks of 3 subjects: \n" +
            sub1 + "\n" sub2 + "\n" +sub3);
    }
```

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```
public static void main (String args []) {
    Student Details sd = new Student Details();
    try {
        sd.getdetails();
    }
  catch (Marks Out of BandException m) {
        System.out.println (m);
    }
}
}
```

Output: -

```
Enter the Student name:-
Raju.
Enter sub1, sub2, sub3 marks:
70
81
102

The Marks of the subject should be less than
100.
Marks OutofBandsException.
```


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Experiment No: 12

Q1) WAP to implement inbuild exceptions.
i.Checked Exceptions(compile time exceptions)
a.ClassNotFoundException
b.IOException

CODE:

```
import java.io.*;
class CheckedExpetion{
 public static void main(String[] args) throws IOException{
//Comment this throws for IOException
System.out.print("Hello World!"); //ClassNotFoundException

 InputStreamReader isr = new InputStreamReader(System.in);
 BufferedReader br = new BufferedReader(isr);
 System.out.print("Try entering String: ");
 String str = br.readLine();
 System.out.print(br); //IOException
```


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```
 }
}
```

OUTPUT:

```
Error: Could not find or load ma
in class CheckedExeption
Caused by: java.lang.ClassNotFou
ndException: CheckedExeption
```


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ii.Unchecked Exception(Run Time Exception)
a.NumberFormatException
b.ArithematicException
c.ArrayIndexOutOFBoundException
d.NullPointerException

CODE:

```
import java.io.*; class
UncheckedException{
 public static void main(String[] args) throws IOException{
try{
 BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));
 System.out.print("Enter a interger: ");
 String str = br.readLine();
int num = Integer.parseInt(str);
 System.out.print(num);
 }
 catch(Exception e){
System.out.println(e);
 }
try{
```


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```
 int c = 4/0;
 System.out.print(c);
 }
 catch(Exception e){
System.out.println(e);
 } try{
int[] a = {1,2};
a[100] = 40;
System.out.print(a);
 }
 catch(Exception e){
System.out.println(e);
 }
 try{
 String str = "Hello";
 str = null;
 int lenght = str.length();
 System.out.print(str);
 }
 catch(Exception e){
System.out.println(e);
 }
 }
```


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OUTPUT:

```
Enter a interger: 4
4java.lang.ArithmeticException: / b
y zero
java.lang.ArrayIndexOutOfBoundsExce
ption: Index 100 out of bounds for
length 2
java.lang.NullPointerException
```


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Q2) Write a Java Program to Create a User Defined Exception Class MarksOutOfBoundsException, if entered marks is greater than 100 or less than zero and then program should create a user defined Exception of type of MarksOutOfBoundsException and musr have a provision to handle it.

CODE:

```
import java.util.*;
class PercentileExpection{ public static void main(String[] args){
 try{
 Scanner input = new Scanner(System.in);
 System.out.print("Enter cet percentile: ");
double pecentile = input.nextDouble();
 checkIfValid(pecentile);
 }
 catch(PercentileOutOfBoundException e){
 System.out.print(e);
 }
 }
 static void checkIfValid(double percentile) throws
PercentileOutOfBoundException{
```


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```
if(percentile<=0.000 || percentile>=100.000){
 throw new PercentileOutOfBoundException(percentile);
 }
 System.out.print("Valid");
 }
}
class PercentileOutOfBoundException extends Exception {
private double detail;
 PercentileOutOfBoundException(double percentile){ detail =
percentile;
 }
 public String ReturnStatement(){
 return "My Exception: "+detail;
 }
}
```

OUTPUT:

```
Enter cet percentile: 103
PercentileOutOfBoundException
=== Code Execution Successful ===
```


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CONCLUSION:-

- **We learnt about exception and error and the implementation of DataInputStream and Buffered reader classes.**
- **We saw the use of try, catch, throw, throws block.**

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